

# ECE 3210 Midterm 2

*Week of: November 3, 2025*

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You have 2.5 hours for 5 problems.

- Show enough (neat) work in the clear spaces on this exam to convince us that you derived, not guessed, your answers.
- Put your final answers in the boxes at the bottom of the page.

You may use two US letter-style size page of notes, front and back.

| Problem            | Score | Possible Points |
|--------------------|-------|-----------------|
| 1                  |       | 20              |
| 2                  |       | 20              |
| 3                  |       | 20              |
| 4                  |       | 20              |
| 5                  |       | 20              |
| <b>Total score</b> |       | 100             |

**1 Short answer**

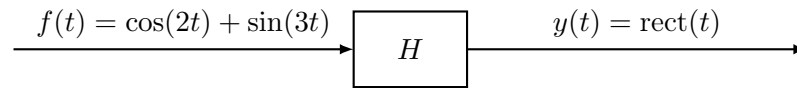
(a) Consider the ODE given below

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y(t) = 2\frac{df}{dt} + 6f(t)$$

with initial conditions  $y(0^-) = 1$  and  $y'(0^-) = 0$  as well as an input  $f(t) = u(t)$ . What is the zero-input response  $y_{zi}(t)$ ?

$y_{zi}(t) =$

- (b) Consider the following system block diagram and the given input/output relationship. Given what we know about this system, can this system be LTI? (Be careful about noticing if things are given in time or frequency domain.)

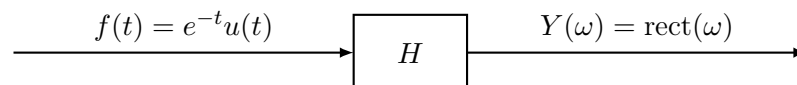


Circle one:

LTI

Not LTI

- (c) Consider the following system block diagram and the given input/output relationship. Given what we know about this system, can this system be LTI? (Be careful about noticing if things are given in time or frequency domain.)



Circle one:

LTI

Not LTI

- (d) Suppose we have signal  $f_1(t)$  that has a bandwidth of 500 Hz and  $f_2(t)$  that has a bandwidth of 1 kHz. If we multiply these two signals together to form a new signal  $f(t) = f_1(t)f_2(t)$ , what the Nyquist sampling frequency (in hertz) for  $f(t)$ ?

$f_s =$

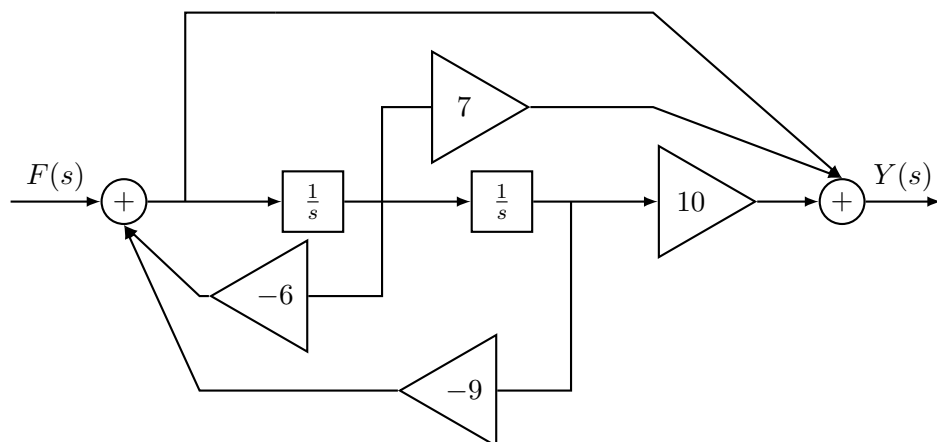
## 2 Block diagrams and transfer functions

- (a) Consider the transfer function

$$H(s) = \frac{s^3 - s + 1}{s^3 + 5s^2 - 3}.$$

Draw the canonical block diagram for this system.

(b) Consider the block diagram below.



(i) What is the transfer function  $H(s)$ ?

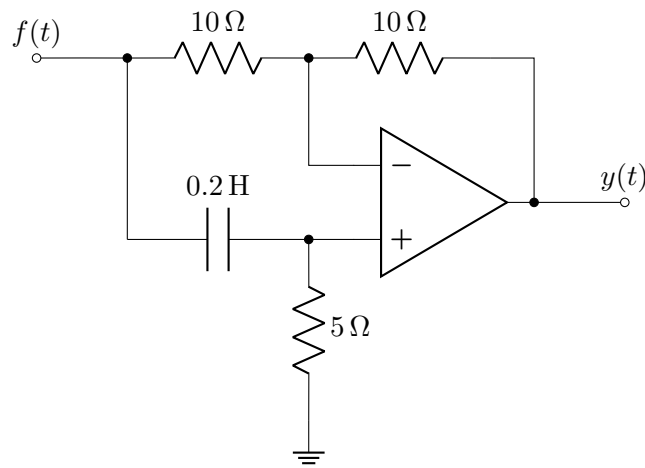
$$H(s) =$$

(ii) What is its impulse response  $h(t)$ ?

$$h(t) =$$

### 3 Circuit analysis and transfer functions

Consider the circuit below where the input is  $f(t)$  and the output is  $y(t)$ . This is the all-pass circuit that we used in the modulation lab. The all-pass system is a system that passes all frequencies equally in magnitude, but changes the phase of different frequencies by different amounts.



- (a) Derive the transfer function  $H(s)$  for this circuit.

$H(s) =$

- (b) Find the impulse response  $h(t)$  for this system.

$$h(t) =$$

- (c) Given an input  $f(t) = u(t)$ , find the zero-state response for  $y(t)$ .

$$y(t) =$$

#### 4 DFT and circular convolution

(a) Consider the sequence

$$x_k = \{1, 0, 1, 0, 1, 0, 1, 0\} \quad \text{for } 0 \leq k \leq 7.$$

Find the 8-point DFT  $X_r$  for  $0 \leq r \leq 7$ . It might be helpful if you can come up with an expression for  $x_k$  in terms of sinusoids or complex exponentials.

$$X_0 =$$

$$X_1 =$$

$$X_2 =$$

$$X_3 =$$

$$X_4 =$$

$$X_5 =$$

$$X_6 =$$

$$X_7 =$$



(b) Consider the following discrete signals

$$f_k = \{2, 1, 0, -1, -2\}$$

$$g_k = \{1, 1, 1, 1\}.$$

(i) Find the linear convolution of  $f_k$  and  $g_k$ .

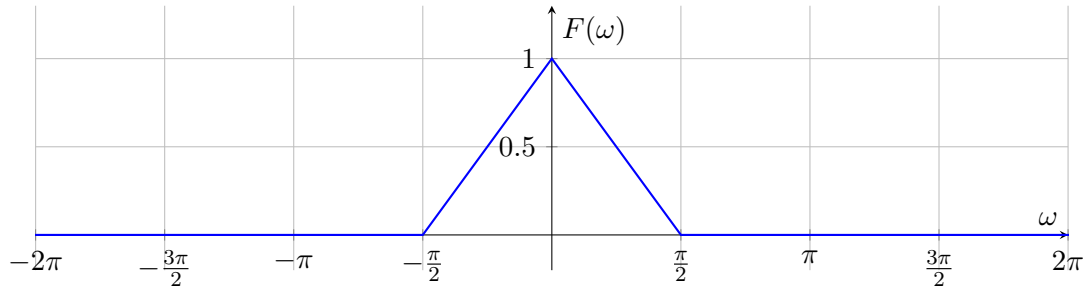
$$y_{k,\text{lin}} =$$

(ii) Find the *5-point circular convolution* of  $f_k$  and  $g_k$ .

$$y_{k,\text{cir}} =$$

## 5 Sampling

Consider the Fourier transform of  $F(\omega)$  of some continuous-time signal  $f(t)$  shown below.

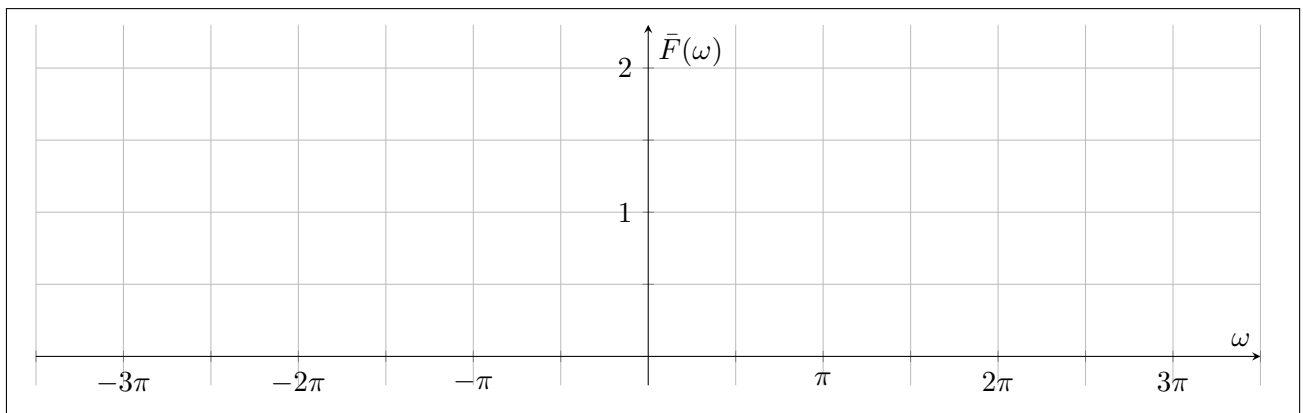
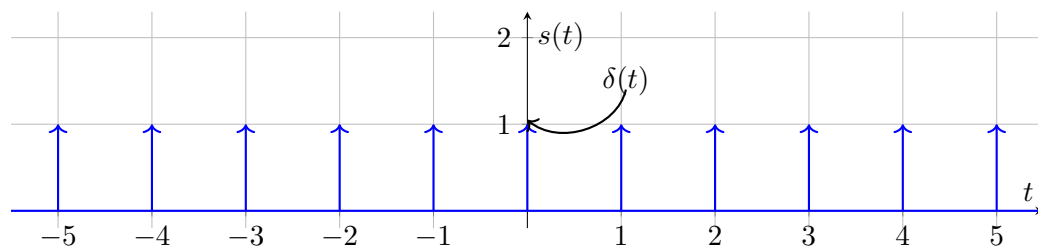


- (a) What is the bandwidth and Nyquist sampling rate ( $f_s$ ) for this signal in hertz?

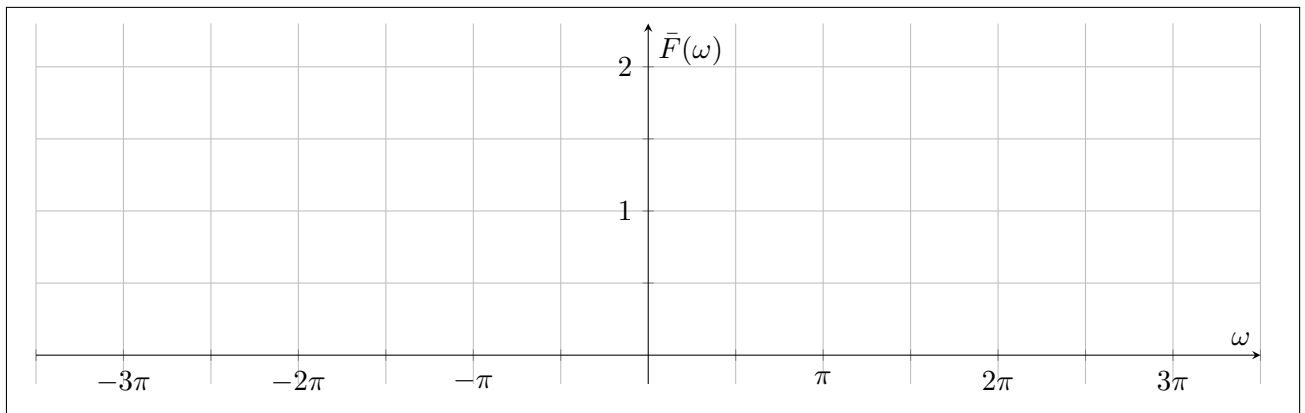
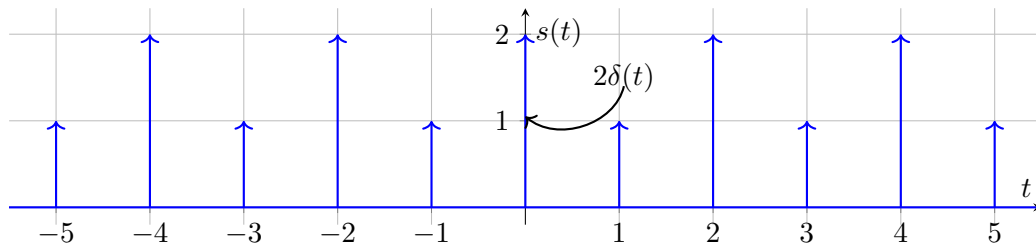
BW=

$f_s =$

- (b) Suppose this signal is sampled using an ideal sampler seen below, sketch the Fourier transform of the resulting signal  $\bar{F}(\omega)$ . These delta pulses are spaced at every 1 second and scaled by 1 (the pulse at  $t = 0$  is annotated in the figure). If we sample with  $s(t)$  such that  $\bar{f}(t) = f(t)s(t)$ , sketch the resulting signal in the frequency domain  $\bar{F}(\omega)$ .



- (c) Now suppose our sampling delta train is messed up and each even numbered delta pulse is scaled by 2 instead of 1 (the pulse at  $t = 0$  is annotated in the figure). If we sample with this new sampling function  $s(t)$  such that  $\bar{f}(t) = f(t)s(t)$ , sketch the Fourier transform of the resulting signal  $\bar{F}(\omega)$ .



- (d) If we are unable to recover the original signal  $f(t)$  from the sampled signal in part (c), what is the minimum cutoff frequency  $\omega_c$  of an ideal lowpass filter that would allow us to recover  $f(t)$ ? If you are unable to recover  $f(t)$ , just write “not possible”.

$\omega_c =$